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RESEARCH ARTICLE

Power to the People! Assigning German Popular Music Genres Based on Listeners' Perspective Using a Genre Tree Approach

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Abstract

Despite the popularity of playlists on streaming platforms, music genres still remain a primary organizational tool among listeners, stakeholders, and researchers. Yet both 'correct' genre labels for individual popular music phenomena and structure of genre taxonomies are often contested. Nevertheless, assumed assignments often form the basis of music genre recognition models in MIR research as well as musical taste and recommender systems research, raising questions about reliability, validity, and comparability of results. By contrast, popular music studies have emphasized the listener-dependent nature of genre assignments. In other words: Genre assignment may be considered a matter of majority voting within a certain community. So why not make a virtue of necessity? We constructed a hierarchical tree from MusicBrainz folksonomy genre votes for 167,762 tracks listened to in Germany across the past 50 years. The resulting Community Music Genre Tree (COMGET) comprises 234 genres—usefully granular but probably too detailed for many research and applied tasks. We address this by an additional tree-folding algorithm that aggregates genres to the desired level of detail for any research problem. For automatic genre assignment of new tracks, we developed classifiers at multiple granularity levels using music features and meta-data drawn from online databases. Results indicate that the approach is able to assign genres that correspond to listeners' perspective, in a transparent, reproducible, and intersubjectively comprehensible way.

Keywords: music genre recognition, hierarchical taxonomy, folksonomy, network analyasis, machine learning classification, German popular music

1. Introduction

Although playlist-driven listening has grown on streaming platforms, genre labels remain the primary way people and institutions organize popular music: Listeners search and browse by genre; playlist editors and DJs assemble genre playlists; A&R, marketing and label teams position and promote artists using genre tags; and music journalists and researchers rely on genre labels to describe and compare popular music (Bonini and Magaudda, 2024; Frith, 1996). Some scholars argue that classificatory practices have weakened since the late twentieth century (DiMaggio, 1987) and that certain genre boundaries are unstable (Silver et al., 2016), but empirical evidence shows that listening and social signaling remain organized around genres (Airoldi et al., 2016; Lonsdale, 2021). On streaming platforms, genre is embedded in metadata and pipeline logic—used as search filters, playlist seeds, and features in recommender systems—which means platform infrastructures amplify genre's practical significance for discovery and market positioning (Valverde, 2025). For researchers, genre labels serve both theoretical and methodological roles: they are variables in preference and reception studies (Lepa et al., 2020), inputs to recommender-system research (Porcaro et al., 2024), and useful aggregators for measuring perceived diversity and mitigating hubness in high-dimensional audio and metadata feature spaces (Flexer et al., 2018; Flexer and Grill, 2016).

However, 'correct' genre labels for individual popular music phenomena (i.e., tracks, artists, releases, scenes, market, playlists, labels, performances etc.) are frequently contested (van Venrooij and Schmutz, 2018). Different actors attach different labels for dif-

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ferent reasons: artists may self-label to narrate their musical biography; publishers and distributors tag releases to support marketing and placement; critics and musicologists apply normative or historical taxonomies; and listeners use genre terms to organize their listening repertoires and to signal identity (Frith, 1996; Negus, 1999). Not only do chosen labels diverge, but the taxonomies they derive from differ as well. Commercial platform taxonomies illustrate these structural differences: The metadata platform *Gracenote* uses a compact set of 25 genre tags (Jiang et al., 2024), whereas *Spotify* exposes many thousands of genre tags (McDonald, nd). Platforms also differ in the level of abstraction at which they apply genre (track, release, or artist) and in assignment strategy (single primary genre versus multi-label tag sets).

These disagreements pose concrete methodological challenges for empirical researchers across MIR and computational musicology. Problems include inconsistent genre labels that degrade generalization of supervised Machine-Learning training (Bogdanov et al., 2019), survey items that ask about genre in ways listeners interpret differently (Lange et al., 2025), and cross-study noncomparability that undermines replication and meta-analysis (Sturm and Flexer, 2023). Faced with these constraints, researchers often pragmatically adopt available commercial or open taxonomies, without systematically engaging with the source of labels or underlying theoretical assumptions (Green et al., 2024). To address this gap, we present a reproducible procedure to derive a scalable and transparently constructed genre taxonomy for a given corpus by analyzing the corresponding genre discourse in a folksonomy. Paired with automated music-genre recognition, this approach can (1) surface empirical regularities and disagreements in label use that inform validity debates (Sturm and Flexer, 2023), and (2) provide a shared, probabilistic 'currency' for cross-disciplinary quantitative work—so that classifiers, preference studies, and sociological analyses operate from a common, well-documented mapping between recordings and meta-genres.

1.1 Musical Genres as hierarchical discourse formation

Musical genres are organized hierarchically: Pachet and Cazaly (2000) identified genealogical links (e.g., *rock* as parent of *alternative rock*), geographical links (*hip hop* as supergenre of *German hip hop*), and historical-period links (*pop* as supergenre of *80s pop*) in existing taxonomies. As Brackett (2016, p. 8) argues, hierarchy depth is not limited to two levels but can extend virtually indefinitely—yet consensus on which labels to select and how to relate them remains contested. Expert-constructed taxonomies show limited agreement (Pachet and Cazaly, 2000), and expert classifications often diverge from listener community norms(Sordo, 2008). Adopting the power-critical perspective of cultural studies (du Gay et al., 2013), we interpret genre discourse through Foucault's (2013) concept of discourse formation, in which industry, listeners, and artists exert unequal influence over which labels circulate and stabilize. Given that emerging genres are primarily defined by listeners (Lena and Peterson, 2008), genre assignment can be understood as implicit majority voting within a community of listeners.

We therefore ground our propsed method in the genre assignments of a large listener community captured in an online folksonomy. Platforms such as *RateYourMusic* (nd), *Discogs* (2024), or *MusicBrainz* (nd) provide crowdsourced 'truth' reflected in aggregate tagging behavior. *MusicBrainz* seems particularly suitable for our endevour: it maintains a detailed whitelist of over 2,000 genre tags without imposing an a priori hierarchy, and its democratic voting system allows users to assign and vote on tags at multiple levels (track, release, artist), enabling nuanced analysis of consensus and disagreement. Although we use *MusicBrainz* as our primary data source, external validation against other folksonomies (i.e., *Discogs*) remains necessary to judge generalizability (Sturm and Flexer, 2023). Schreiber (2015, 2016) demonstrated that hierarchical genre relationships can be inferred from tag co-occurrence patterns on digital platforms. Crucially, however, genre assignment conventions vary across listener communities in different countries and time periods (Pichl et al., 2017; Nie, 2022). As a case study, we thus anchor our exemplary analysis in popular music tracks listened to in Germany over the past 50 years to ensure the resulting taxonomy reflects a coherent discursive context.

For the use of hierarchical genre taxonomies in empirical research, three pragmatic issues arise. First, balanced genre distributions improve statistical power and classifier performance (Japkowicz and Stephen, 2002). However, subgenres are typically assigned to fewer tracks than their super categories (Bogdanov et al., 2019), creating inherent imbalance that must be considered when constructing or aggregating within the taxonomy. Second, the optimal number of genres depends on the research question: surveys of musical taste may require moderate to high resolution (Lange et al., 2025, e.g., 24 genres), robust machine learning models often perform better with fewer, broader categories (Genussov and Cohen, 2010), and fine-grained subcultural analyses may demand more genres (Caparrini et al., 2020). At the same time, preserving consistency with the overall tree structure is desired. This demands a scalable taxonomy that supports multiple resolutions derived from a single underlying hierarchy. Third, applying such taxonomies to large corpora in computational contexts requires an automated classifier that can assign genres in an efficient, consistent and transparent way, serving as a neutral arbiter when human dissent over labels is expected.

1.2 Towards an automated classifier for popular music studies

Music genre recognition is a core task in Music Information Retrieval. Green et al. (2024) provide a systematic state-of-the-art review and critique how genre theory from the social sciences and humanities is often overlooked when selecting features (for a systematic review on genre theories see Merlini, 2020). Commonly, researchers in MIR and music psychology assume that genre can be inferred from audio features alone (Tzanetakis and Cook, 2002; Rentfrow et al., 2011). This conflicts with sociological genre theories that describe genres as "systems of orientations, expectations and conventions that circulate between industry, text and subject" (Neale, 1980, p. 19), emphasizing extra-musical context alongside sound. Franco Fabbri's (1982) genre theory offers a synthesis: musical style (instrumentation, rhythm, harmony, timbre) and release milieu (subcultural context, artist positioning, market placement) together inform how listeners assign genres. Modern MIR tools can extract style features from audio signals (Bogdanov et al., 2013), and open music databases provide access to distributional metadata (label, chart performance, platform placement).

Lyrics occupy an intermediate position, conveying both emotional expression as stylistic content (Brand et al., 2019) and contextual information through language codes and topics that signal subcultural affiliation (Carbone et al., 2024). A classifier might need to pay special attention to the *mainstream pop* or *pop* metagenre that Fabbri (1982) describes as encompassing eclectic, commercially optimized works that resist association with a specific milieu. In addition, artists from former niche scenes are often retrospectively relabeled as *pop* or *mainstream* after achieving commercial success (Holt, 2007, p. 20). Therefore, popularity itself seems to be a relevant feature dimension for classification. We thus propose a feature space spanning four dimensions—musical style, release context, lyrical content, and popularity—to operationalize the multi-dimensional constitution of genre.

Drawing on the theoretical and methodological foundations outlined above, we pose three research questions:

RQ1 How can we construct an empirically grounded and balanced hierarchical genre tree for popular music in Germany?

RQ2 How can we aggregate to a scalable number of supergenres from the inferred tree suitable for machine learning and statistical applications?

RQ3 Can an automatic classifier predict genres at different levels of detail with robust performance using stylistic, distributional, lyrical, and popularity features?

In the following sections, we will: (2) introduce a method to create a balanced hierarchical genre taxonomy based on genre co-occurrences in folksonomy tags, (3) describe a folding algorithm to reduce the number of genres for various tasks, (4) present automatic classifiers for different levels of granularity, and (5) discuss our findings and summarize the results.

2. RQ1: Balanced Hierarchical Genre Tree

2.1 Data – Popular Tracks in Germany (POPTRAG)

We grounded the analysis on a curated corpus of 254,334 popular tracks listened to in Germany over the past 50 years (POPTRAG). Four sources comprise this corpus: 215,702 tracks from weekly *MediaControl* single and album charts (18 Nov 1977–21 Nov 2024; 75% linking success to *Spotify* identifiers); 26,949 *Spotify* recommendations from 126 German-market genre seeds (retrieved 29 Nov 2024; median 325 per seed, IQR 25, range 49–343); 8,801 tracks from 100 featured *Spotify* playlists for the German market (median 90 tracks per playlist, IQR 42, range 28–750); and 11,994 entries from *Germany's Daily Top 200 Spotify* charts (01 Jan 2017–30 Nov 2024). We excluded tracks shorter than 30 seconds (853 tracks, likely non-musical snippets) and retained only unique title-artist combinations, preferring earlier release dates for duplicates.

To determine usable *genre* labels, we applied conservative filtering: we discarded tracks lacking *genre* tags or carrying only non-musical labels, removed tags absent from the *MusicBrainz* whitelist (snapshot 16 Jun 2025), required tags to appear across at least 50 distinct artists to ensure community support, and retained only tracks with at least two tag votes. Excluded non-musical tags comprised *non-music*, *interview*, *comedy*, *nature sounds*, *children's music*, *spoken word*, and *standup comedy*. These criteria removed 86,572 tracks (34%), yielding a *MusicBrainz*-tagged subset of $N =$ 167,762 tracks annotated with 234 distinct music genre tags spanning release years 1924–2024 (median 2012, IQR 16 years). When track-level tags were missing, we fell back to release tags, then artist tags, resulting in: track tags for 90,470 tracks (54%), release tags for 5,915 tracks (4%), and artist tags for 71,377 tracks (43%). Users assigned between 1 and 22 different tags per track (median 4, IQR 3) and cast between 2 and 135 votes per track (median 6, IQR 7). This multi-source, vote-weighted corpus provided a broad, listener-centered basis for inferring genre co-occurrences.

2.2 Methods: From Co-Occurrences to Taxonomy

We built a hierarchical genre taxonomy from *MusicBrainz* tag co-occurrences by adapting (Schreiber, 2015) to make every transformation explicit and reproducible. We first constructed a weighted co-occurrence network where nodes represented *genre* labels and directed edges $e_{A \to B}$ quantified how strongly *genre A* predicted *genre B* on the same track. For each ordered pair (A, B), we computed a vote-weighted conditional frequency:

$$e_{A \to B} = P(B \mid A) \times \frac{1}{|T_{A,B}|} \sum_{t \in T_{A,B}} \frac{v_{t,B}}{v_{t,\text{total}}} \tag{1}$$

where $T_{A,B}$ denotes tracks tagged A and B, $v_{t,B}$ the votes for B on track t, and $v_{t,\text{total}}$ the total votes on t. This formula produced asymmetric edges ($e_{A \to B} \neq e_{B \to A}$) that combined directional co-occurrence with community vote strength. To illustrate: suppose 1,000 tracks carried *metal* and 600 also carried *rock*, giving $P(rock|metal) = 0.6$. If the average vote share for *rock* on co-tagged tracks equalled 0.5, then $e_{metal \to rock} = 0.6 \times 0.5 = 0.3$. Conversely, 6,000 tracks carried *rock*, 600 of those carried *metal*, so $P(metal|rock) = 0.1$; with an average *metal* vote share of 0.7 on *rock* tracks, $e_{rock \to metal} = 0.07$. These weights express both prevalence and tag confidence.

Second, we resolved reciprocal links to obtain a unidirectional scaffold. For each pair, we subtracted the weaker directional signal from the stronger and kept only the positive residual:

$$e'_{A \to B} = \left(e_{A \to B} - e_{B \to A}\right)_+ \tag{2}$$
$$e'_{B \to A} = \left(e_{B \to A} - e_{A \to B}\right)_+ \tag{3}$$

In the example this yields $e'_{metal \to rock} = 0.6 - 0.07 = 0.53$ and $e'_{rock \to metal} = 0$, making *metal* → *rock* a clear candidate supergenre link.

Third, we enforced a strict tree by assigning each node its single strongest outgoing edge as its *supergenre*; all other edges were discarded. Components without *supergenre* were attached to an artificial root *POPULAR MUSIC* to guarantee connectivity. In the example, *metal* would attach to *rock* unless a stronger edge existed, and *rock* would attach to the root if it had no *supergenre*.

Finally, we balanced the tree across levels. We estimated each *genre's size* as the sum of its assignment probabilities across tracks, computed the Gini coefficient within each level, and defined overall imbalance as the weighted mean of level Ginis. We searched for the least-imbalanced tree by iteratively removing the weakest hierarchical link, reattaching the resulting dangling tree to *POPULAR MUSIC*, and recomputing imbalance; we selected the iteration with minimal overall imbalance.

2.3 Results: Community Music Genre Tree (COMGET)

The initial POPTRAG *genre* co-occurrence network contains 234 *genre* nodes forming a single connected component with 45.14% density, indicating relatively high inter-tag connectivity. By median, *genres* co-occurred with 97 different others (IQR 63.5; range 3 for *waltz*

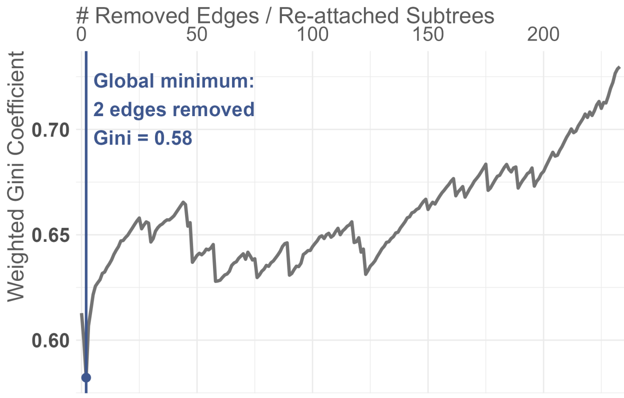


Figure 1: Weighted Gini coefficient (0–1) across iterations of cutting weak genre links and reconnecting dangling subtrees to *POPULAR MUSIC*.

Level	# nodes	# tracks	Gini
1	3	total: 47,846 (29%) 12,472; 16,401; and 18,973	0.136
2	119	total: 90,381 (54%) median: 263 IQR: 789 range:10.4 to 5832	0.643
3	110	total: 29,343 (17%) median: 156 IQR: 259 range:14.7 to 1496	0.531
4	2	total: 192 (0.1%) 55.6 and 136	0.420

Table 1: Structure of COMGET at different hierarchy levels; track counts are sums of genre tag assignment probabilities in POPTRAG.

to 229 for *pop*), revealing large differences between niche and ubiquitous tags. After resolving reciprocal directions, each *genre* received on average 29.5 incoming subgenre links (IQR 32; min 0; max 228 for *pop rock*) and 55 outgoing supergenre links (IQR 32; min 0 for *rock*; max 105 for *dub*).

The balance optimization (Figure 1) reattached *pop* and *hip hop* directly to *POPULAR MUSIC*—*pop* had been subordinate to *rock*, and *hip hop* to *pop*. The three weakest remaining links are *electronic* to *pop* (0.22), *classical* to *pop* (0.27), and *jazz* to *pop* (0.29). The final *Community Music Genre Tree* (COMGET) is highly leaf-heavy: 89.36% of nodes lack children. *Rock* emerges as the largest *supergenre* with 63 direct *subgenres*, and only *rock*, *pop*, and *hip hop* remain connected directly to *POPULAR MUSIC*, forming three broad meta-categories. The tree exhibits four hierarchy levels (excluding the artificial root); see Table 1 for structural details. An interactive visualization is available in the digital appendix.

3. RQ2: Scalable Granularity

3.1 Methods: Folding Algorithm

We designed a greedy algorithm to fold COMGET into a scalable number of *supergenres* suitable for diverse research questions and statistical applications. The folded tree must satisfy three requirements: optimal balance, a user-selectable number of supergenres, and sufficient track counts per genre for robust machine learning and statistical analysis. The algorithm iteratively merges the least populous *genres* furthest from the root with their *supergenres* until all *genres* meet a minimum track count threshold. *Genre size* was again computed as the sum of assignment probabilities across all tracks. We conducted a grid search across minimum *genre sizes* from 1,000 to 15,000 in increments of 5, maximizing within-level balance as measured by the weighted Gini coefficient. For a desired number of *supergenres*, users can now select the folding solution corresponding to the local Gini minimum within any given target range. Local minima across the full function identified statistically optimal solutions regarding balance. Please refer to the digital appendix for a more detailed description of the algorithm.

For external validity assessment, we compared COMGET-12 genre assignments to the *Discogs* folksonomy *genre* labels using a contingency table. To enable this comparison, we assigned each track a single label via majority voting: we mapped all valid *MusicBrainz* tags assigned to a track to their *supergenres* in folded COMGET, then assigned the track the *supergenre* tag with the highest summed votes. Ties were broken by selecting the *genre* with lower total assignment probability in the full dataset, favoring more detailed *genre* assignments.

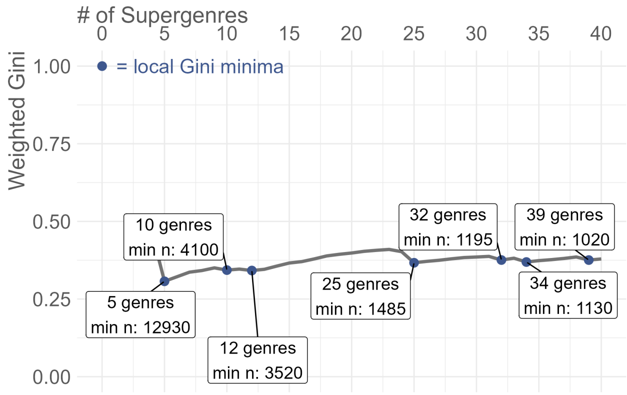


Figure 2: Weighted Gini coefficient (0–1) across folding solutions for minimum track counts from 1,000 to 15,000 in increments of 5.

3.2 Results: Germany's Popular Music Supergenres

The folding algorithm identified multiple optimal solutions for COMGET (Figure 2): a low-resolution solution with 5 *supergenres* (min. 12,930 tracks, Gini 0.307), medium-resolution solutions with 10 and 12 *supergenres* (min. 4,100 and 3,520 tracks; Gini 0.343 and 0.342), a high-resolution solution with 25 *supergenres* (min. 1,485 tracks, Gini 0.367), and finer solutions with 32, 34, and 38 *supergenres* (min. 1,195, 1,130, 1,020 tracks; Gini 0.376, 0.369, 0.375). We focused subsequent analyses on solutions with 5, 12, 25, and 32 *genres*, representing different granularities suitable for various research questions (Figure 3).

COMGET-5 comprises *rock*, *pop*, *hip hop*, *electronic*, and *metal* (in descending order of track count). In COMGET-12, *jazz* and *schlager* emerge as distinct *pop subgenres*, while *rock subgenres* including *heavy metal*, *alternative rock*, *pop rock*, *hard rock*, and *indie rock* appear at the same level. Notably, listeners do not classify *heavy metal* as a *subgenre* of *metal* but as a direct *subgenre* of *rock* on the same hierarchy level as *metal*.

COMGET-25 introduces further *rock subgenres* (*punk*, *progressive rock*, *folk*, *folk rock*, *classic rock*, *blues rock*, *singer-songwriter*) and additional *pop subgenres* (*soul*, *classical*, *synth-pop*, *r&b*). A third hierarchy level appears with *death metal* as a *subgenre* of *metal* and *power metal* as a *subgenre* of *heavy metal*. COMGET-32 adds *house* as an *electronic subgenre* and additional *rock* and *pop* branches (*blues*, *country*, *new wave*, *soft rock*, *dance-pop*, *reggae*). *hip hop subgenres* appear only in more fine-grained solutions (*pop rap* as 33rd genre, *trap* as 38th genre).

Discogs' 15-genre taxonomy approximates COMGET-12 in scale. For approximately 5% of tracks with COMGET assignments, we found no *Discogs* counterpart, introducing minimal popularity bias. The contingency table (Figure 4) reveals strong alignment for *electronic*, *hard rock*, and *metal*. Even where *Discogs* includes a *Jazz* category, COMGET's *jazz* overlaps substantially with *Discogs*' *Electronic* and *Pop*.

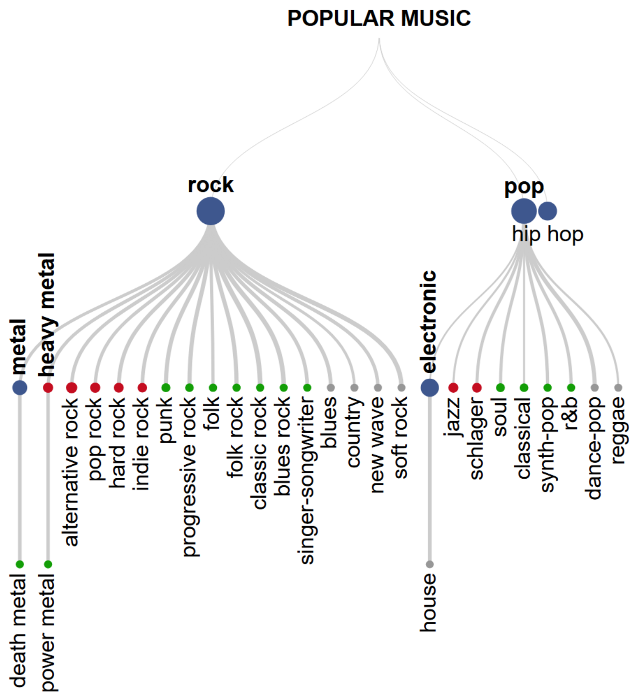


Figure 3: Folded COMGET hierarchies at four resolutions (5, 12, 25, 32 supergenres). Colors indicate genres introduced at each finer resolution (blue = 5; red = 12; green = 25; grey = 32). Node size represents track count in POPTRAG (sum of probabilities).

Discogs

COMGET-12	Blues	Brass & Military	Children's	Classical	Electronic	Folk, World, & Country	Funk / Soul	Hip Hop	Jazz	Latin	Non-Music	Pop	Reggae	Rock	Stage & Screen
alternative rock	0	0	0	0	10	0	0	1	0	0	0	1	0	88	0
electronic	0	0	0	1	93	0	0	1	1	0	0	1	0	3	0
hard rock	0	0	0	0	0	0	0	0	0	0	0	0	0	99	0
heavy metal	0	0	0	0	1	0	0	0	0	0	0	0	0	99	0
hip hop	0	0	0	0	8	0	0	90	0	0	0	1	0	1	0
indie rock	0	0	0	0	10	1	0	0	0	0	0	2	0	85	0
jazz	0	0	0	1	11	1	2	2	71	1	0	8	0	4	0
metal	0	0	0	0	3	0	0	0	0	0	0	0	0	97	0
pop	0	0	0	3	30	0	10	7	3	3	0	20	2	22	0
pop rock	0	0	0	0	8	0	0	1	1	0	0	12	0	77	0
rock	1	0	0	0	7	2	0	1	2	0	0	2	0	84	0
schlager	0	0	1	0	4	2	0	0	0	0	1	76	0	16	0

75% 50% 25% 0%

Figure 4: Contingency table comparing COMGET-12 and Discogs genre assignments; values show $P(\text{Discogs} \mid \text{COMGET-12})$.

Similarly, *pop rock* distributes across Discogs' *Rock*, *Pop*, and *Electronic*; *schlager* concentrates in *Pop* (similar label to its COMGET *supergenre*) with minor overlap to *Rock*. Five *Discogs* categories—*Blues*, *Brass & Military*, *Children's*, *Non-music*, and *Stage & Screen*—show minimal correspondence to any COMGET-12 *genre*, mostly consistent with our noise-reduction filtering and focus on popular music. COMGET's *pop* distributes widely across *Discogs genres* (*eEectronic*, *Pop*, *Rock*), reflecting pop's eclectic character. Seven of twelve COMGET-12 *genres* overlap most strongly with *Discogs' Rock*, suggesting *Discogs* lacks granular rock subgenre labels needed to represent German popular music adequately.

4. RQ3: ML-Classifier

4.1 Methods: Outcome, Features, Workflow

To determine whether an automated classifier can predict the *supergenres* identified by the folding algorithm, we derived *single genre tags* for each track using majority voting at four levels of detail (COMGET-5, COMGET-12, COMGET-25, COMGET-32) and trained independent classifiers for each. We enriched POPTRAG with 225 variables spanning *musical style*, *release context*, *lyrical content*, and *popularity* from *Spotify*, *MusicBrainz*, *AcousticBrainz*, *Genius*, and *Deezer* via fuzzy matching and web scraping. *AcousticBrainz* provided community-calculated *audio features*; *MusicBrainz* supplied artist demographics (*persona*, *origin*, *birth year*, *gender*); *Spotify* and *Deezer* contributed *popularity* metrics and *Echo-Nest audio features*. From *Spotify's artist-level genre* tags, we derived 58 *distribution genre* flags (e.g., "Is a track's artist tagged as 'reggae' on *Spotify*"?) using the same co-occurrence algorithm applied to COMGET, but operating on *Spotify artist genres* as input. *Distribution genres* capture how the music industry positions and markets artists within commercial taxonomies, which may diverge from listener perception due to strategic positioning, legacy labelings, or cross-genre appeal (Valverde, 2025). By including these industry-assigned categories alongside audio and lyrical features, we provide the classifier with a signal of release milieu that complements community-based genre assignments and tests whether industry framing influences how listeners ultimately categorize music.

We first evaluated four learners on a 20% subsample at low and medium resolution to balance prediction performance and interpretability: elastic net (GLMNET), regularized discriminant analysis (RDA), random forest (RF), and light gradient boosted machine (LightGBM). For each, we tuned key hyperparameters: GLMNET (*penalty*, *mixture*), RDA (*covariance fraction* and *identity fraction*), RF (*split variables*, *leaf node size*, *tree depth*; *number of trees*: 500 for tuning, 1,000 for final models), and LightGBM (*boosting rounds*, *learning rate*, *leaf count*, *tree depth*, *feature fraction*).

In our workflow, we first discarded observations with more than 40% missing values, retaining 155,093 tracks (92%). We performed an 80/20 train-holdout split at the artist level to mitigate artist effects (Flexer and Schnitzer, 2010), and used 5-fold cross-validation (also stratified-split on artist level) for tuning. We lumped categorical levels with fewer than 100 observations in any fold into an "other" category, then discarded variables that contained only this category. All categorical predictors were dummy-coded with the most common level as reference.

We tuned hyperparameters using the macro-averaged F1 score (mean per-class F1, set to 0 for unpredicted classes) to treat all *genres* equally during evaluation. Preprocessing was applied in sequence: median imputation, adaptive class balancing (excluded from holdout set), zero-variance removal, and normalization. Adaptive balancing used a single tunable *target-ratio* parameter governing rebalancing degree. The algorithm computed a reference count from class frequencies (median for ≤ 10 classes; 60th percentile for 11–20 classes; 65th percentile for > 20 classes), then downsampled classes exceeding reference × *target ratio* using NearMiss (Zhang and Mani, 2003) and upsampled classes below reference ÷ *target ratio* using Borderline SMOTE (Han et al., 2005) or regular SMOTE when sufficient danger observations were unavailable. This approach preserved more data than fixed-ratio sampling while remaining tunable: *target ratio* = 1 enforced perfect balance; higher values retained original imbalance. The reference-count heuristic scaled to problem complexity, preventing excessive downsampling at higher resolutions.

We employed Bayesian optimization (Snoek et al., 2012) with a Latin hypercube initial grid (20 points for full analysis, 10 for prototyping) (Husslage et al., 2011), a maximum of 50 Bayes steps (25 for prototyping), early stopping if no improvement occurred after 10 iterations, and exploration of uncertain hyperparameter regions after 5 iterations without improvement. We selected hyperparameters using the one-standard-error rule (Hastie et al., 2009), trained final models on the full training set, and evaluated on the holdout set. *Community certainty* served as a reference point for evaluating final classifier performance, since models cannot exceed the consistency of the underlying community assignments. *Certainty* was computed as the mean assignment probability across votes for each track assigned to the *single-label genre*. All random operations used seed 42 for reproducibility.

4.2 Results: Performance in the Light of Uncertainty

Model comparison (Figure 5) shows that all four learners perform similarly on the 20% subsample in terms of macro F1. LightGBM emerged as the strongest performer at both resolutions (COMGET-5 and COMGET-12) and was selected as the final learner for all subsequent analyses. GLMNET ranked second for COMGET-5 but performed worst for COMGET-12, indicating that its linear decision boundary assumption does not scale to higher-resolution classification tasks.

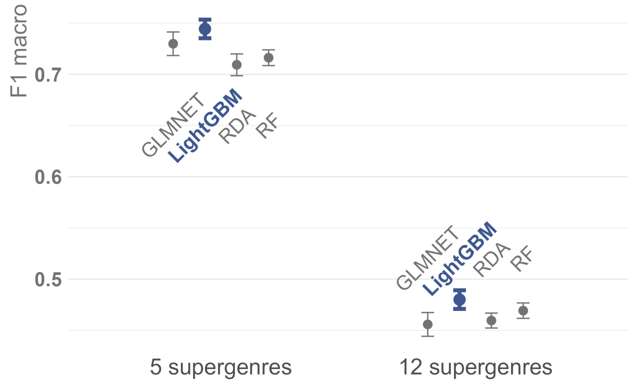


Figure 5: Performance comparison of different learners (GLMNET, RDA, RF, GBM) on 20% of the data set for COMGET-5 (left) and COMGET-12 (right); plots show mean and standard error of $F1_{macro}$ across CV folds.

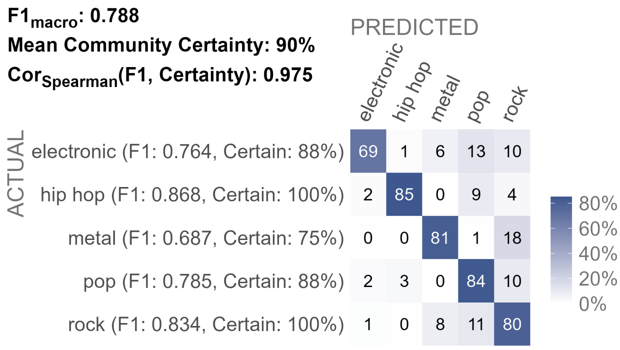
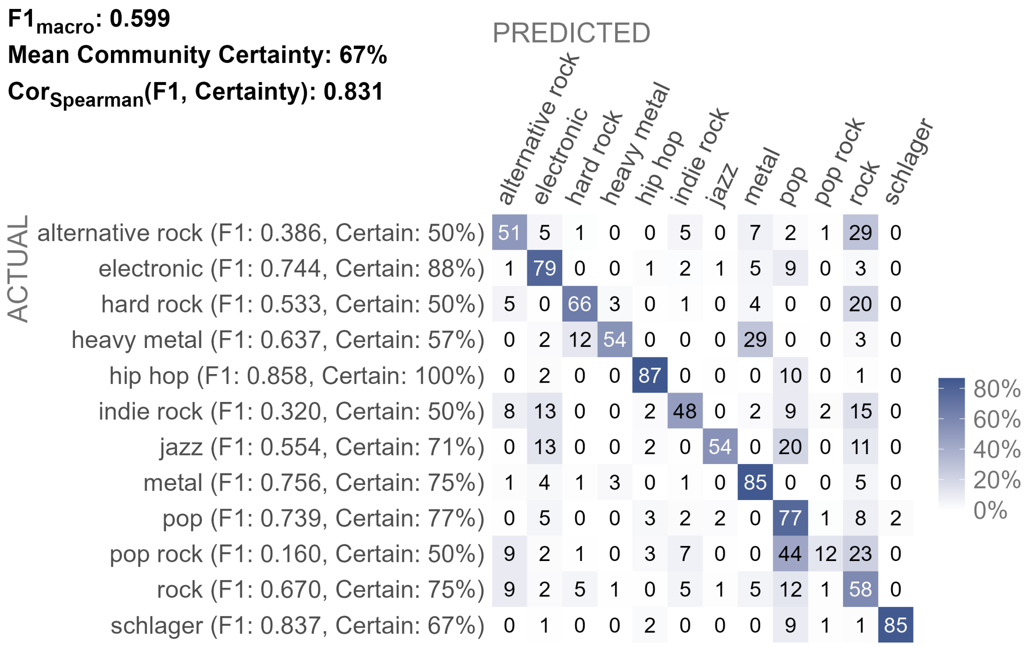


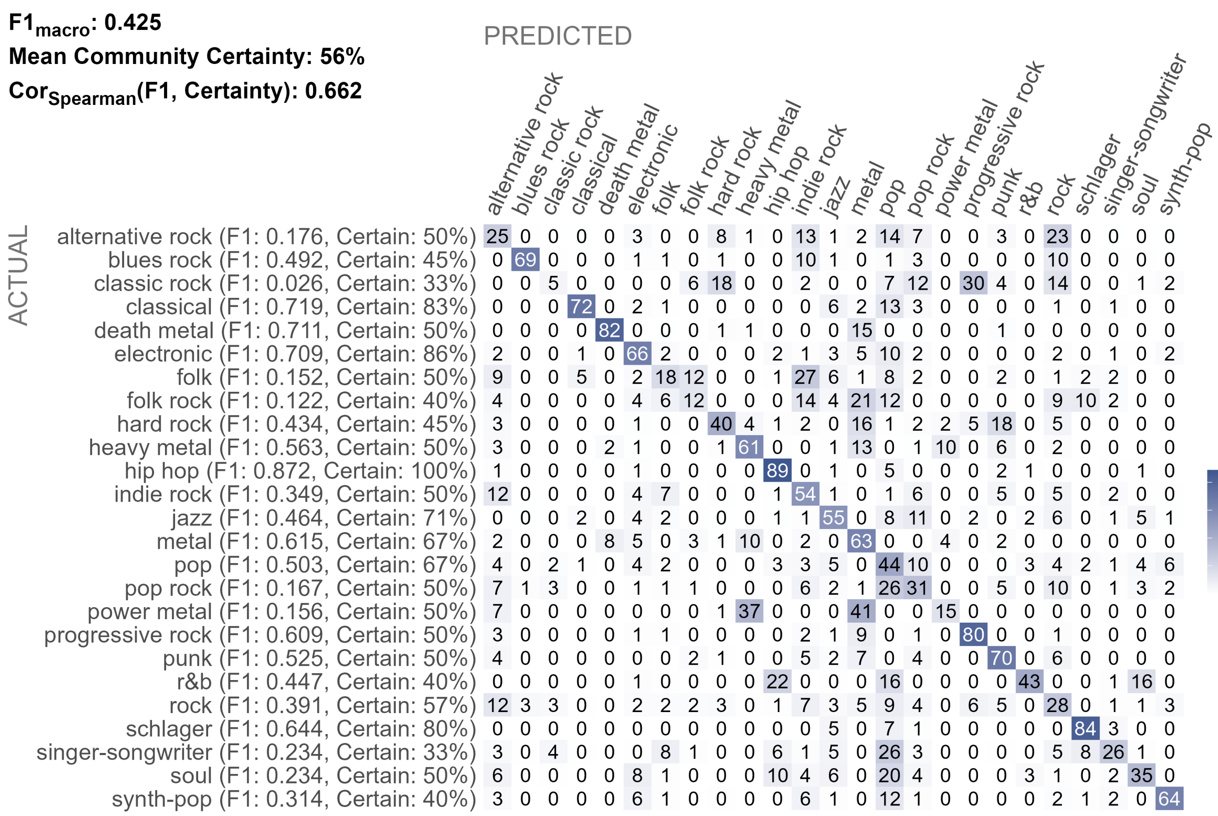
Figure 6: Confusion matrix for the final GBM model on the hold-out set for COMGET-5. Values show $P(\text{Predicted}|\text{Actual})$; *certainty* indicates the mean assignment probability for each *genre* across community votes.

The final GBM models on the full dataset (Table 2) achieved $F1_{macro} = 0.803$ (SE: 0.008) on CV folds and 0.788 on the hold-out set for COMGET-5, indicating slight overfitting. Per-class F1 correlates strongly with *community assignment certainty* ($Cor_{Spearman} = 0.975$). Given mean *community certainty* of 90%, the classifier performs satisfactorily. Cohen's $\kappa = 0.723$ on the hold-out set ($\kappa_{CV} = 0.734 \pm 0.009$) meets established inter-rater agreement standards in the social sciences (Hildebrandt et al., 2015, p. 327).The confusion matrix (Figure 6) reveals the model struggles most with *electronic*, often confusing it with *pop* and *rock*. Electronic music's stylistic elements frequently appear across pop and rock subgenres, explaining the observed misclassifications. Similarly, *metal* misclassifications with *rock* reflect their close relationship.

COMGET-12 performance degraded to $F1_{macro;\ CV} =$



(a) COMGET-12



(b) COMGET-25

Figure 7: Confusion matrices for the final GBM models on the hold-out set. Values show *P*(Predicted|Actual); *certainty* indicates the mean assignment probability for each *genre* across community votes.

# genres	boosting rounds	tree depth	feature fraction	leaf count	learning rate	target ratio
5	854	4	83	98	0.049	1.177
12	846	10	35	4	0.037	1.106
25	898	5	48	67	0.006	2.394
32	854	4	83	98	0.049	2.462

Table 2: Hyperparameter settings for final LightGBM models at different granularity levels.

0.572 ± 0.013 and $\mathrm{F1}_{\text{macro; hold-out}} = 0.599$ ($\kappa_{\mathrm{CV}} = 0.618 \pm 0.010$; $\kappa_{\text{hold-out}} = 0.623$), with lower *community certainty* (67%), indicating greater difficulty for both community and classifier. Per-class F1 and *certainty* remain highly correlated ($\mathrm{Cor}_{\mathrm{Spearman}} = 0.831$). The confusion matrix (Figure 7a) shows primary confusion among *rock subgenres* (*alternative rock*, *hard rock*, *indie rock*, *pop rock*). *Heavy metal* frequently confuses with *metal* but rarely vice versa, and sometimes with *hard rock*, despite being recognized as distinct from *rock*. *Indie rock* and *jazz* often misclassify as *electronic*; *jazz* also confuses with *pop*. *pop rock* performs worst (F1 = 0.160), matching its low *community certainty* (50%). It confuses primarily with *rock* (F1 = 0.670, certainty = 75%) and *pop* (F1 = 0.739, certainty = 77%), suggesting *pop rock* shares ambiguous characteristics with both. Conversely, *schlager* and *hip hop* classify well (F1 = 0.837 and 0.858, respectively) despite different *community certainties* (67% and 100%), indicating these *genres* possess distinct characteristics in the feature space.

COMGET-25 showed further degradation ($\mathrm{F1}_{\text{macro; CV}} = 0.419 \pm 0.005$; $\mathrm{F1}_{\text{macro; hold-out}} = 0.425$; $\kappa_{\mathrm{CV}} = 0.486 \pm 0.010$; $\kappa_{\text{hold-out}} = 0.469$) alongside decreased *community certainty* (56%). The correlation between *certainty* and F1 weakened ($\mathrm{Cor}_{\mathrm{Spearman}} = 0.662$), yet remained high, suggesting other factors influence performance at higher granularity. The confusion matrix (Figure 7b) again reveals persistent *rock subgenre* confusion (*alternative rock*, *blues rock*, *classic rock*, *folk rock*, *pop rock*) and *metal*-related confusion (*death metal*, *folk rock*, *hard rock*, *heavy metal*, *power metal*). A secondary confusion center emerges around *pop*, which frequently misclassifies with *alternative rock*, *classical*, *electronic*, *folk rock*, *pop rock*, *r&b*, *singer-songwriter*, *soul*, and *synth-pop*. The model performs well (F1 > 0.7) on *classical*, *death metal*, *electronic*, and *hip hop—genres* with distinct feature signatures and moderate to high *community certainty* (ranging 50%-100%). Poor performers (*alternative rock*, *classic rock*, *folk*, *folk rock*, *pop rock*, *power metal* with F1 ranging 0.026-0.176, *certainty* ranging 33%-50%) represent fuzzy *genres* difficult to classify for both community and model.

COMGET-32 yielded similar degradation ($\mathrm{F1}_{\text{macro; CV}} = 0.400 \pm 0.003$; $\mathrm{F1}_{\text{macro; hold-out}} = 0.403$; $\kappa_{\mathrm{CV}} = 0.477 \pm 0.005$; $\kappa_{\text{hold-out}} = 0.486$) with decreased *certainty* (54%). Importantly, the correlation between *certainty* and F1 recovered notably ($\mathrm{Cor}_{\mathrm{Spearman}} = 0.795$), indicating *community uncertainty* becomes the dominant performance factor at extreme granularity. Given similarity to COMGET-25, detailed confusion matrix results appear in the digital appendix.

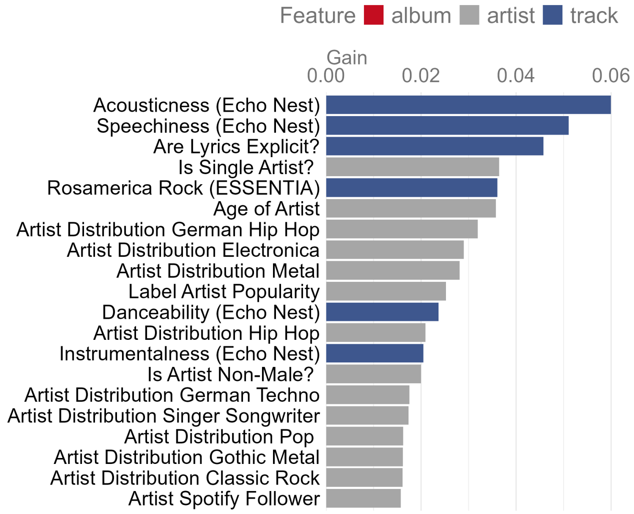
Figure 8 shows the top 20 most important features for automated classification of COMGET-5, COMGET-12, and COMGET-25 with the final LightGBM learner. While we find similar features in the top 20 across all three levels of detail (e.g., *acousticness (ECHONEST)*, *artist distributed as "metal"*, *age of artist*), the importance of these features differs between the levels of detail. For COMGET-5, features that describe general style characteristics (e.g., *acousticness*, *speeciness*, *explicitness of lyrics*) and artist persona (e.g., *single artist vs. band*, *age of artist*) are most important. For COMGET-12, *popularity* features and the *release year* gain importance. For COMEGET-25 the most important features are mostly artist-related, indicating that fine-grained *genre* distinctions rely primarily on release-milieu signals rather than audio properties.

5. Discussion

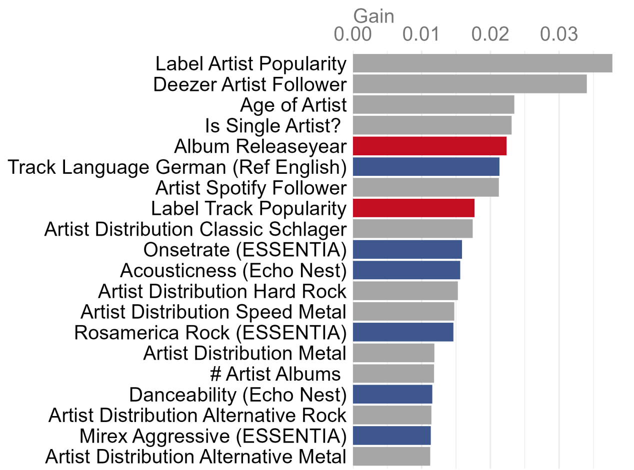
The aim of our study was to propose a methodologically transparent procedure for constructing hierarchical genre taxonomies from listener-community discourse. We demonstrate three key contributions: (1) a balanced hierarchical genre tree derived from folksonomy tag co-occurrences without manual curation, (2) a scalable folding algorithm that generates optimally balanced solutions at user-specified granularities, and (3) automated classifiers that predict genres with performance bounded by community assignment certainty. We instantiate this approach on German popular music using *MusicBrainz* folksonomy votes and the POPTRAG corpus, yielding COMGET: a transparent, reproducible, and empirically grounded alternative to commercial or expert-curated taxonomies.

5.1 Balanced and Scalable Representation of Discourse

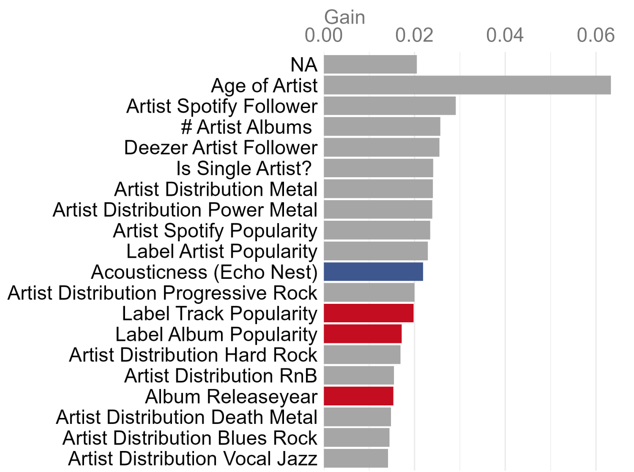
We adapted Schreiber's transparent algorithm (Schreiber, 2015) to construct hierarchical genre taxonomies but grounded our labels exclusively in folksonomies rather than mixing commercial taxonomies with community votes. This choice reflects a power-critical perspective from cultural studies (du Gay et al., 2013) that is often adapted in popular music studies. Our method thus engages both MIR methodology and humanities debates about cultural classification (DiMaggio, 1987). Democracy operates at multiple levels: genre labels and co-



(a) COMGET-5



(b) COMGET-12



(c) COMGET-25

Figure 8: Top 20 most important features for automated classification at three granularity levels. Values show relative feature importance from the final LightGBM models.

occurrences rely on listener votes, and our algorithmic pipeline embeds democratic decision-making throughout; we selected the strongest hierarchical link for each supergenre assignment and employed majority voting to convert multi-label tags into single labels for classifier training.

Furthermore, our approach adheres directly to requirements for the use of genres in empirical environments. The method generates balanced hierarchies without manual thresholds or hyperparameters, making it portable to any corpus with multi-tag genres. We demonstrated this when inferring distribution genres from *Spotify* artist tags using POPTRAG. Crucially, our folding algorithm derives optimally balanced solutions at different scales using the same statistical criterion (weighted Gini coefficient) that balanced the full taxonomy, enabling researchers to work at various levels of detail while maintaining a consistent conceptual framework. Beyond researchers, this approach is also of interest for industry stakeholders such as radio stations and streaming services needing to organize large music collections at multiple granularities.

We would like to emphasize that the COMGET taxonomy does not represent historical genealogy as Pachet and Cazaly (2000)conceptualized it. Rather, it depicts how listeners hierarchically organize German popular music discourse, i.e., which genres they relate to one another. Genealogical links such as *rock* as supergenre of *alternative rock* appear, but so do counterintuitive ones: *classical* and *jazz* emerge as *pop* subgenres. This reflects POPTRAG's composition. POPTRAG mostly comprises accessible and exceptionally popular tracks, e.g. from crossover artists, popular opera arias, soundtracks, popular jazz singers, or smooth jazz. Combined with the very frequent use of broad labels (Bogdanov et al., 2019) this leads to the observed hierarchy. Similarly, both *heavy metal* and *metal* appear as direct *rock* subgenres, explained by heavy metal's roots in late-1970s psychedelic rock and hard rock alongside metal's later distinctiveness (Gracyk, 2016).

Rock, pop, and hip hop emerged as three primary genre clusters in German popular music. These findings coincide with Silver et al.'s (2016) obervation of three main genre clusters within international popular music (based on 3 million artists on *MySpace*, 2007). In line with their observations, Hip hop stands apart: it shows minimal subgenre differentiation, strong coherence around its supergenre, and fewer subgenres than rock or pop clusters. However, while Silver et al. (2016) distinguished between "Rock' n Roll" and "Niche" genres, we find that rock genres form a distinct cluster in popular German music, while pop genres are diverse in terms of musical styles, awith niche genres scattered across all three clusters.

5.2 Classifier Performance in MGR

As (Holt, 2007, p. 5) noted, semantic codes of musical genres are fragile. This makes MGR such a challenging endeavor. Even experts in the field do not entirely agree on genre assignments (P'almason et al., 2018); so apparent misclassifications might also reflect genuine genre overlap rather than model failure. In other words: Models cannot exceed human consistency (Sturm and Flexer, 2023). Given community uncertainty in genre assignments, our classifiers achieved reasonable overall performance. We observe the expected degradation with increasing label granularity (Genussov and Cohen, 2010), mirroring declining community certainty. Also, subgenres are harder to be detected than supergenres in general, probably due to shifting boundaries (Silver et al., 2016). Artist-level features dominated across granularities, indicating that release milieu (i.e., how platforms and industry position artists) guides listener genre perception. This aligns with sociological genre theory emphasizing industry, text, and subject as coconstitutive (Neale, 1980). Stylistic features, such as audio and lyrics features, contributed significantly but secondarily. Lyric features especially showed weak predictive power, despite utility in other work (Mayer et al., 2008). Incorporating LLM-based lyric embeddings might capture cultural context more richly and better discriminate release milieu (Prevedello et al., 2024).

5.3 Limitations and Future Research

Several constraints limit generalization. *MusicBrainz* launched in 2000 without individual vote timestamps, so COMGET aggregates roughly 25 years of listener votes rather than reflecting a single moment; genre boundaries and perception likely shifted within this window. Platform user demographics and expertise remain opaque (approximately 1.2 million anonymous editor accounts; MusicBrainz nd), preventing assessment of demographic or expertise bias. Popularity bias affected coverage: 36% of initial tracks received fewer than two MusicBrainz votes, potentially underrepresenting niche or rare recordings. However, since we target Germany's popular music genres, this constitutes only a minor limitation. Methodologically, collinearity among metric predictors complicates feature-importance interpretation and may have caused instability in which features appeared predictive across models.

Future work should pursue three directions. First, convene popular music experts to review the taxonomy, label exemplar tracks, and identify misassignments to strengthen COMGET's validity. Second, replicate the pipeline on independent folksonomies or regional corpora of tracks (e.g., US and UK charts) to reveal how national listening cultures reshape genre structure. Third, enhance classifier prediction through advanced lyric features or learners.

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